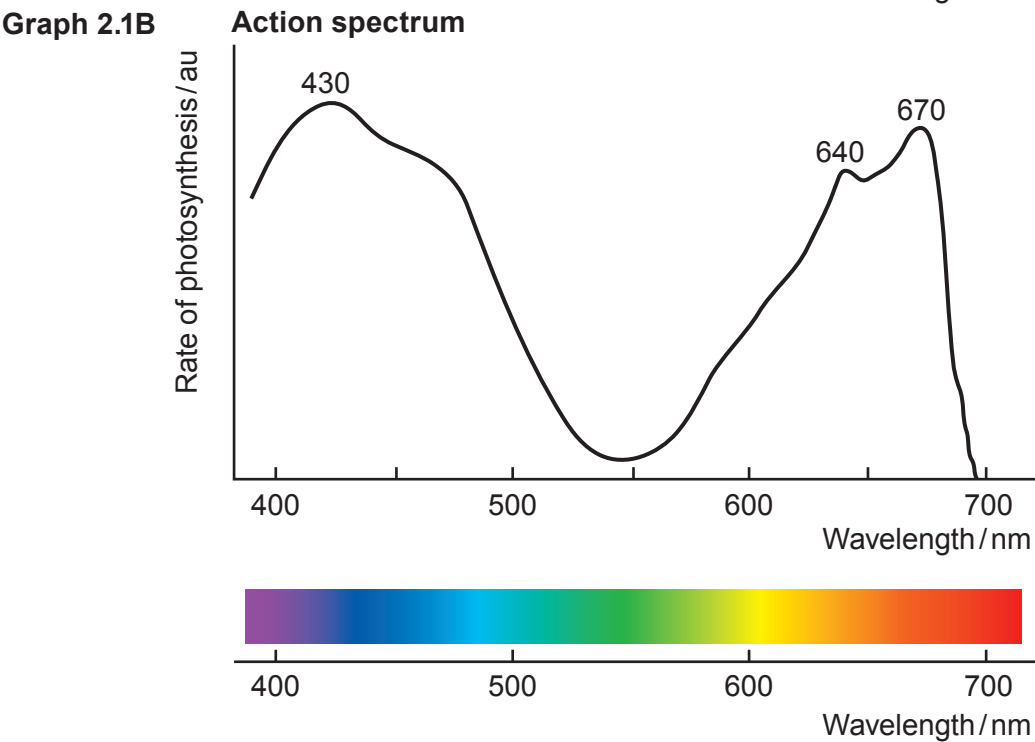
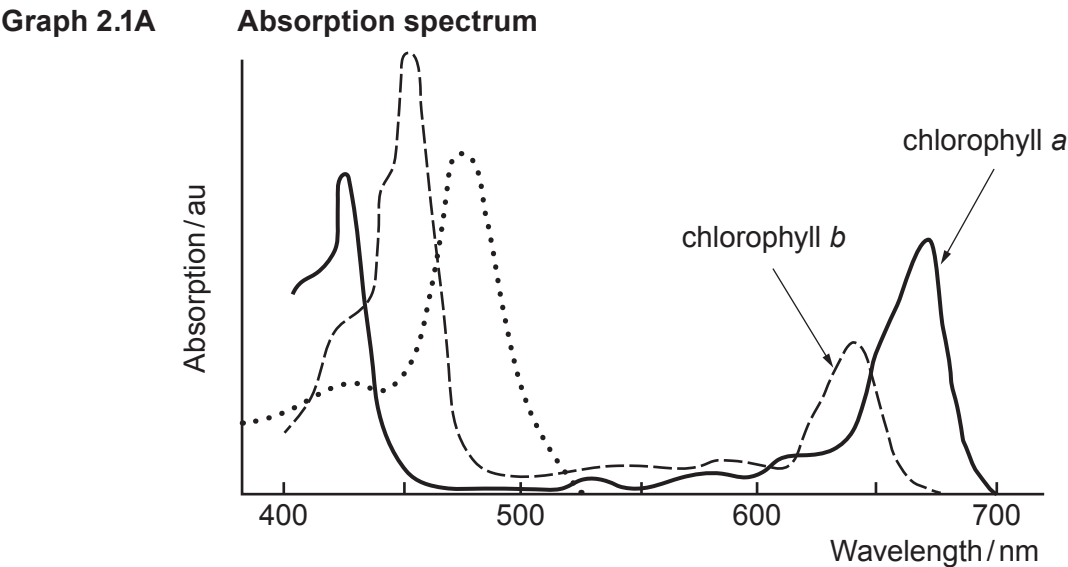


2. Most terrestrial plants use chlorophylls *a* and *b* to construct pigment-protein complexes which harvest light. **Graph 2.1A** shows an absorption spectrum for a terrestrial plant and the **Graph 2.1B** shows corresponding action spectrum.



(a) Describe the relationship between the absorption spectrum and the action spectrum and state a suitable conclusion which explains the relationship. [2]

.....

.....

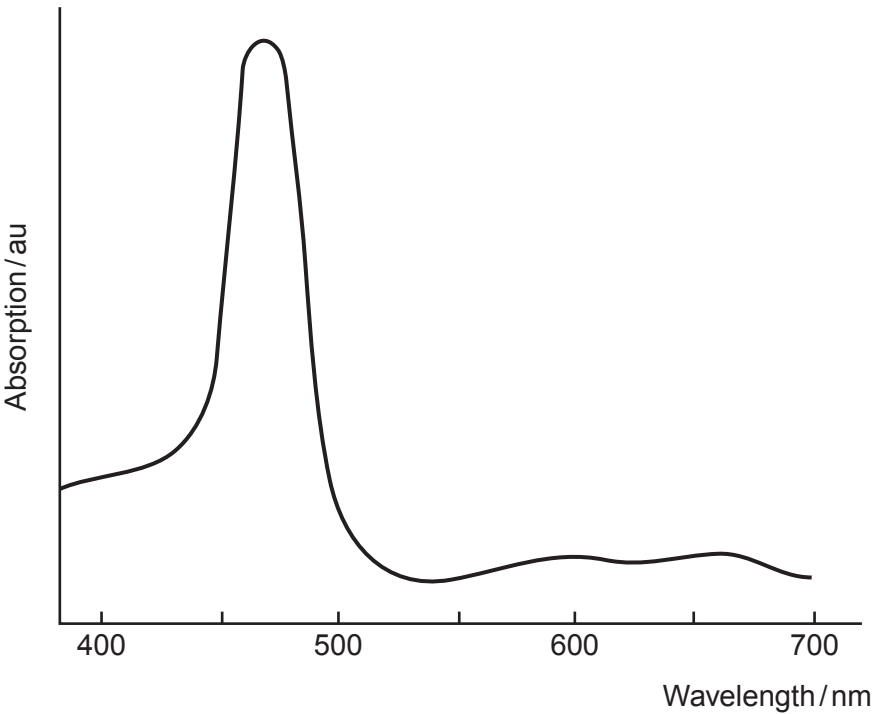
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- (b) Different taxa contain different photosynthetic pigments. Diatoms are aquatic photosynthetic organisms. Their chloroplasts contain chlorophyll *a*, but instead of chlorophyll *b* they contain chlorophyll *c*. **Graph 2.2** shows the absorption spectrum of chlorophyll *c*.

Graph 2.2

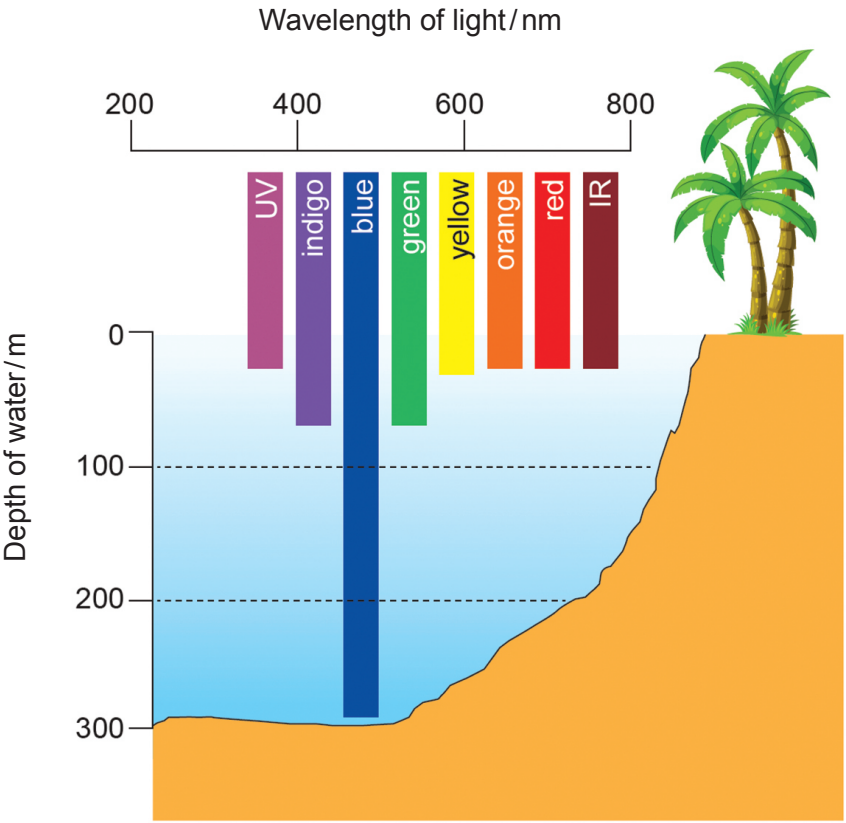


- (i) Describe the main difference between the absorption spectrum for chlorophyll *b* and the absorption spectrum for chlorophyll *c*. [1]
-
- (ii) With reference to the peaks labelled on the action spectrum in **Graph 2.1B**, predict how the action spectrum for diatoms would differ from that of terrestrial plants. [1]
-



Image 2.3 shows the depths that different wavelengths of light are able to penetrate water.

Image 2.3



- (c) Use information from **Graph 2.2** and **Image 2.3** to explain why diatoms living at a depth of 200m have chlorophyll c instead of chlorophyll b. [1]

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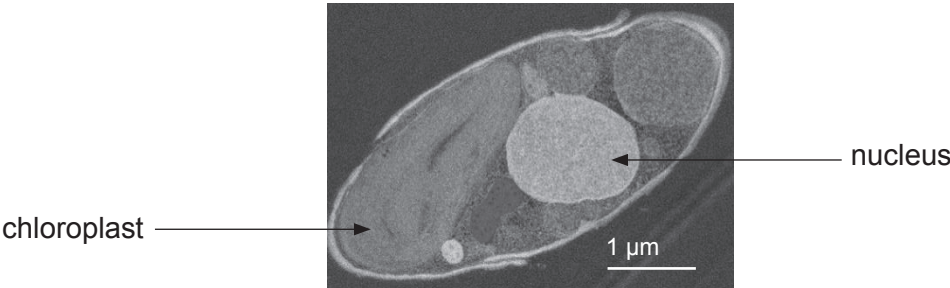
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Image 2.4 shows an electronmicrograph of a single diatom.

Image 2.4



- (d) Using information from **Image 2.4**, classify diatoms into their Domain. Give a reason for your choice. [2]

Domain

Reason

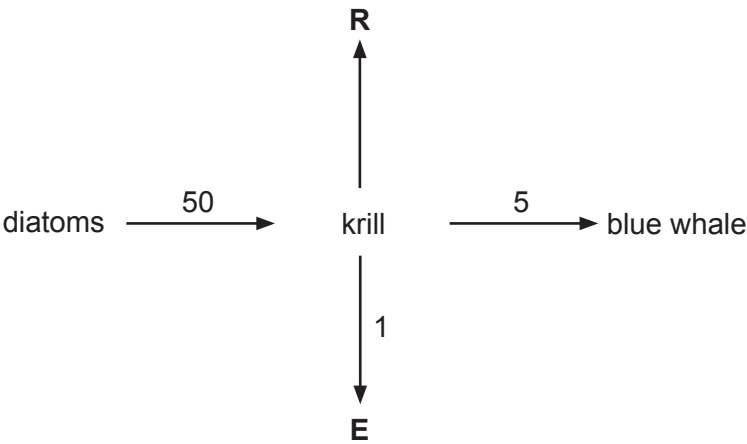
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- (e) Diatoms are responsible for about 40% of marine productivity and because the oceans cover about 70% of the Earth’s surface they make a great contribution to global productivity. Blue whales feed on tiny crustaceans called krill, by filtering seawater through sheets in their mouths called baleen. Krill feed on diatoms.

Net primary productivity (NPP) for diatoms has been estimated to be $50 \text{ g m}^{-3} \text{ day}^{-1}$ and secondary productivity for krill has been estimated to be $5 \text{ g m}^{-3} \text{ day}^{-1}$. **Image 2.5** shows a simplified food chain, the numbers shown are in $\text{g m}^{-3} \text{ day}^{-1}$. **R** represents respiration and **E** represents excretion.

Image 2.5



- (i) Calculate the rate at which krill use diatom biomass for respiration (**R**). [1]

Rate = $\text{g m}^{-3} \text{ day}^{-1}$

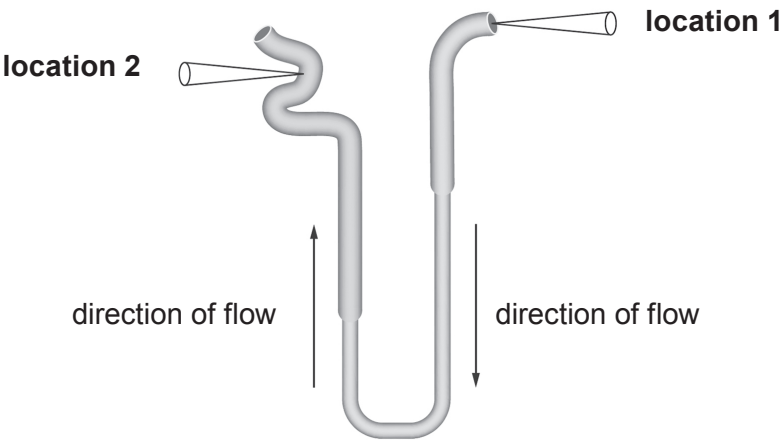
- (ii) It is estimated that a single blue whale needs to consume 8000 kg of biomass per day. The biomass of krill was estimated as 25 g m^{-3} . Calculate the volume of water a whale needs to filter per day to take in 8000 kg of biomass. **Give your answer in standard form.** [3]

Volume of water = $\text{m}^3 \text{ day}^{-1}$



A third experiment was carried out to study how structure **X**, shown in **Image 5.1**, is involved in the formation of urine. **Image 5.4** shows regions of tubule on either side of structure **X**. Samples were withdrawn from two locations, **1** and **2**. The concentration of sodium ions (Na^+) was measured in each location.

Image 5.4



(c) (i) Explain why these locations were chosen for sampling. [1]

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(ii) The Na^+ concentration in the sample taken from **location 1** was higher than the sample taken from **location 2**. Use your knowledge of the function of both limbs of the structure in **Image 5.4** to explain this result. [4]

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